

Stage 1: Defining the Problem

Identifying stakeholders and their needs

	Justification	Needs
Child	Primary users of the chair and will be spending most time in the chair. If they do not like the design, the chair will go unused.	Expressed: The chair should suit children's proportions, offer proper back support and be overall comfortable to sit in . Threshold: As one of their few possessions, chair must withstand frequent use and rough play. Threshold: The ability to support a child's weight is a basic, non-negotiable need for functionality. Latent: With limited resources in relief shelters, the chair must be engaging and act as a source for play.
Parent	Will be responsible for upkeep and maintenance of the chair as well as ensuring the chair is accessible for their child.	Expressed: The chair should be easy to set up - ideally it should take no longer than 45 seconds. It should also have storage underneath the seat. Threshold: The chair must be easy to move but also have a sturdy build to withstand transporting (i.e. with one hand), and must be suitably safe and accessible for their child to use.
Distributor	Must be practical for NGOs to distribute as well as affordable.	Expressed: The chair should be able to be assembled/disassembled into a flat sheet of cardboard for purposes of transit/storage, and the material should be cost effective yet strong. Extraneous spending is impractical as funding will be minimal.
Volunteer	Volunteers handle large scale setup in relief centers. Limited training or tools available.	Expressed: Design should be simple with minimal components so that assembly is efficient (10s to 45s). Expressed: The simple design instructions should be identifiable, clear and self-explanatory.
Manufacturer	Must be able to produce at scale efficiently, with minimal materials, waste, and simple cutting.	Expressed: Production does not require special equipment and tools, machine's capabilities are limited to cutting. Latent: The manufacturing of the chair is not time consuming so that manufacturing and distribution can happen on short notice as natural disaster is unpredictable.

Table 1: Stakeholders and their Needs

Problem Background:

Natural disasters often force large groups of people into temporary, emergency housing, often in school gymnasiums and public buildings. In these environments, resources are scarce and each person has few, if any, personal items. Children in these environments are stressed and scared. So, in an effort to reduce the trauma these children experience, a NGO has proposed a design competition to create a cardboard chair that serves as fun and furniture for children in emergency shelters.

Identified Target Design Specifications, based on needs identified in Table 1.

1. The chair is made entirely out of cardboard.
2. The chair will support a minimum weight of 60lbs (more is preferred).
3. The flat unfolded chair form will span a rectangle of with an area less than 15,484 square cm (less is preferred).
4. The assembly instructions are printed on each chair and are visual.
5. The chair can be assembled by 2 people in less than 45 seconds (less time is preferred).
6. The height of the seat off the ground must be between 26 cm and 32cm.
7. The seat will have dimensions of 36 cm x 36 cm.
8. The chair must include a backrest with height between 30 cm and 40 cm.
9. The chair contains at least one built in game (more is preferred) that contributes towards child engagement.
10. The aesthetics of the chair are child friendly and welcoming.
11. The chairs will have a customisable name tag.
12. The chair will have an internal storage capacity, that is completely covered from outside view, of 24 cm x 24 cm x 23cm.

Stage 1

Stage 2: Generating Ideas

At the beginning of stage 2, 12 rounds of C-sketch were conducted. Presented below are 6 of the 12 which met the most number of design specifications.

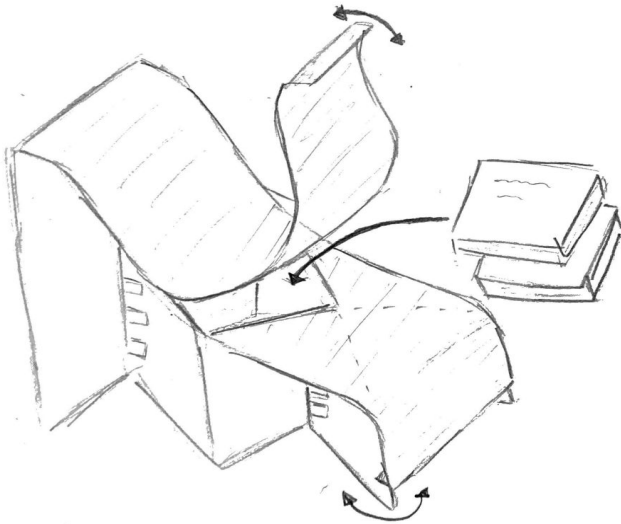


Figure 1: X Base Chair Sketch

Strengths:

- Kid friendly and visually appealing
- Large storage space

Weakness:

- Curve seat surface means poor distribution of weight in testing

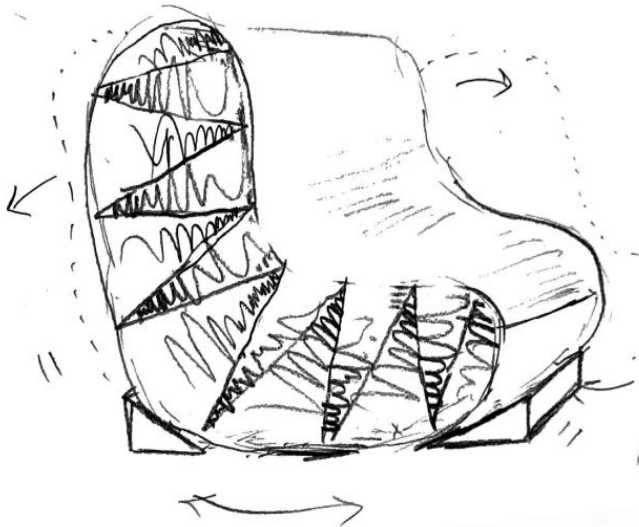


Figure 2: Rocking Chair Sketch

Strengths:

- Kid friendly and visually appealing
- Large storage space

Weakness:

- Lots of cardboard required in W-folds

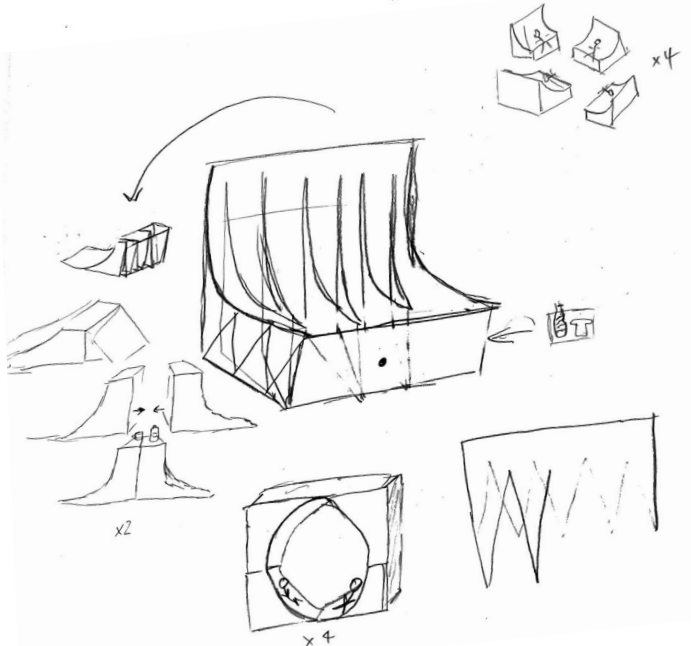


Figure 3: Curved-Chair Sketch

Strengths:

- Comfortable
- Visually appealing

Weakness:

- Lack of storage
- The curved surface may reduce the sturdiness of the chair

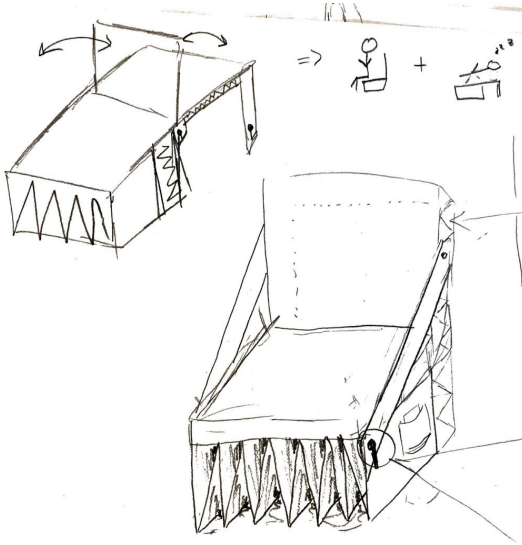


Figure 4: Convertible Bed/Chair Sketch

Strengths:

- Creative and multifunctional
- Medium storage space

Weakness:

- May exceed unfolded size
- Complex design may take more than 45s to assemble

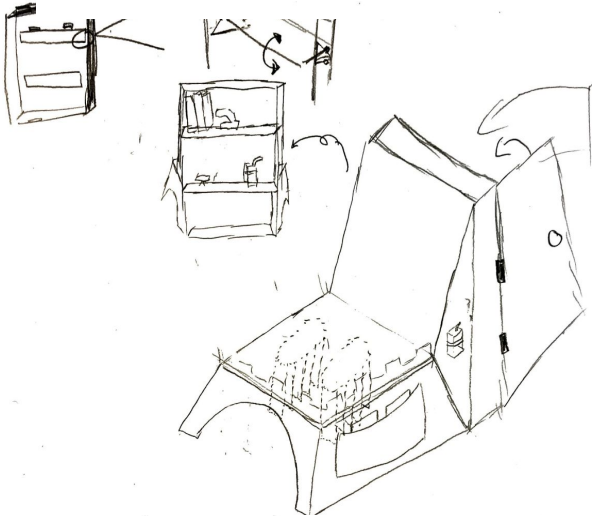


Figure 5: Shelving Chair Sketch

Strengths:

- Practical with built-in storage
- Good stability

Weakness:

- Shelving may cause slow assembly
- Lots of cardboard required

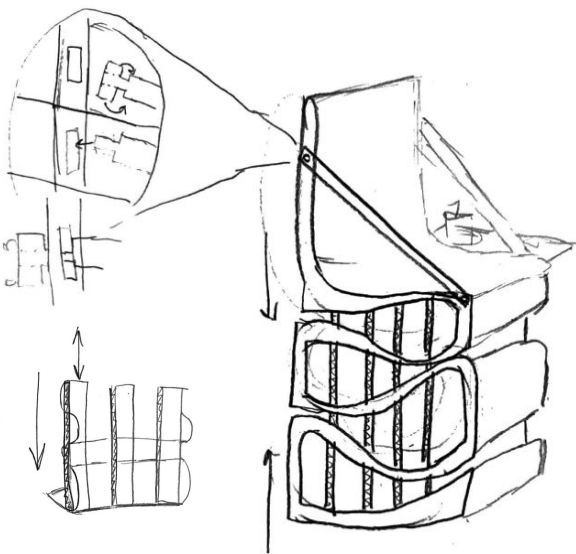


Figure 6: Grid-base Chair Sketch

Strengths:

- Unique look and creative design

Weakness:

- May exceed maximum cardboard amount
- Lack of storage space
- Unconventional shape may not maximize effective cardboard usage

Stage 2

Stage 3: Identifying Most Promising Solution

Screening

After two rounds of C-sketch and playing around with cardstock models, we ended up with 12 different designs, which we then screened down to six that met all the target design specifications (with some iteration required)

Ranking Method: Borda Count

Six designs were carried through to ranking. Each member ranked all six of the designs, with their first choice of design receiving 5 points, second receiving 4 points all the way to the last ranked design which received 0 points. This method prevented fixation from individual group members, as well as limited group biases as voting was anonymous. Ultimately, three of the highest ranking designs were carried through to scoring: the “X Base Chair”, “Triangle Chair”, and “Three Piece Chair”.

Through the use of comprehensive physical scale models, we could test the assembly, storage capacity and durability of the chosen chair design. This allowed us to recognise the flaws of each chair, in order to pull elements from other designs and make improvements to the scoring on the evaluation criterion.



Figure 7: Ranked Voting results for Chair Designs

Scoring

We scored the top three designs against the evaluation criteria, and assigned points to each design based on their estimated performance of the scaled models within each criterion.

As the Triangle Arm and X base chair both scored very high, our final design combines the basic structure of X base design with the rest of the Triangle Arm design. The main improvements for the triangle arm chair were its aesthetic elements, entertainment factor for kids, and its load capacity/stability. These were all iterations from elements of the X base design.

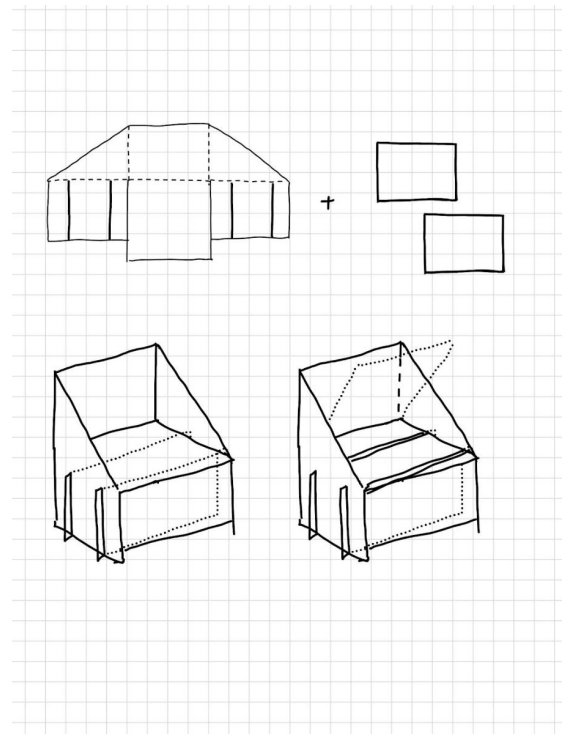


Figure 8: Initial Sketch of the Triangle Armrest design

	Area	Time	'Fun'	Aesthetic	Storage	Load	Comfort	Ease of use	Total
Triangle Chair	10	4	2	4	10	6	3	7	46
X Base Chair	1	0	8	8	10	8	7	3	45
Three Piece Chair	6	5	2	6	5	6	5	3	32

Table 2: Estimated Scoring of Selected Chairs

*Same 36 cm x 36 cm seat dimensions for all three designs

Stage 3

Stage 4: Develop and Test Solution

After constructing and testing prototypes with different loads, we realised the horizontal support pieces under the seat could not support much weight. We also noticed these structures severely limited storage capacity (as seen in scoring), so we sought to modify the design by incorporating other ideas.

Iteration:

- We incorporated the durability and storage maximisation of the “X base” chair while keeping the small area footprint and folding structure of the “Triangle Arm” chair.
- We cut the X base square in half to reduce material, but found this unacceptably decreased the storage capacity and durability of the support structures. Instead, we split the “X base” support structure into two triangular portions to increase the angle of the support.
- We included handles at either side of the armrests to improve mobility and comfort of our chair.

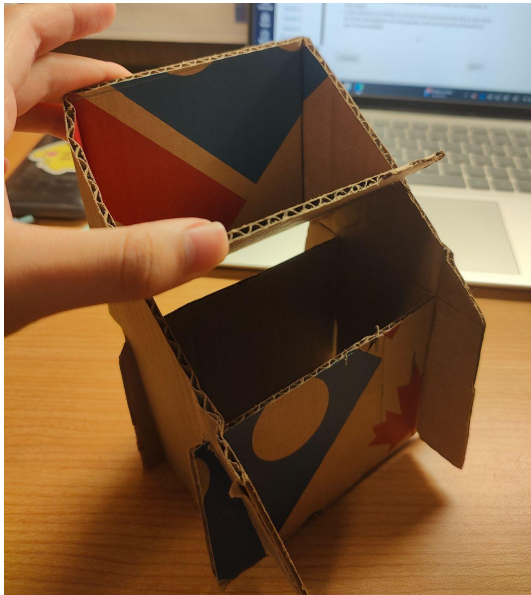


Figure 9: Demonstrating flip-up seat of Triangle Armrest design

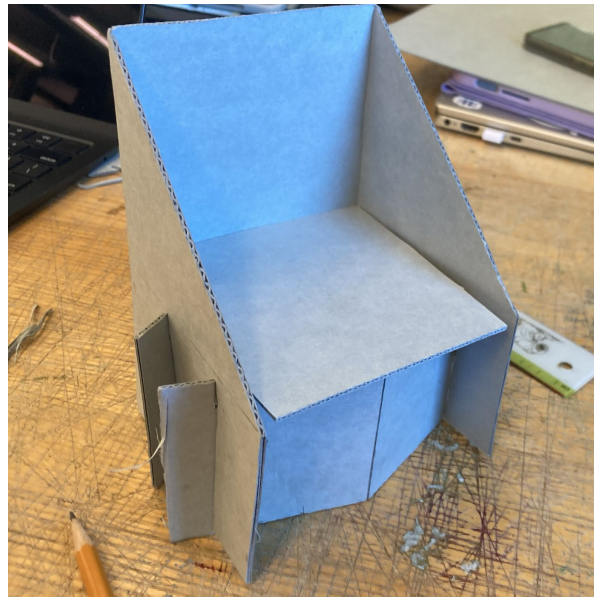


Figure 10: Rigid paperboard clean prototype of triangle armrest chair

The final design of our chair includes the most useful features we identified from our generation and refinement stages. The finishing touches on this design include consistent critter theming sure to delight any child, and games painted onto the chair (tic-tac-toe and ring toss when the chair is placed upside down) to inspire children with how to play with the chair.

Stage 5: Implementing Solution

Comfortable backrest Accessible handles

Customizable name tag Supportive armrests

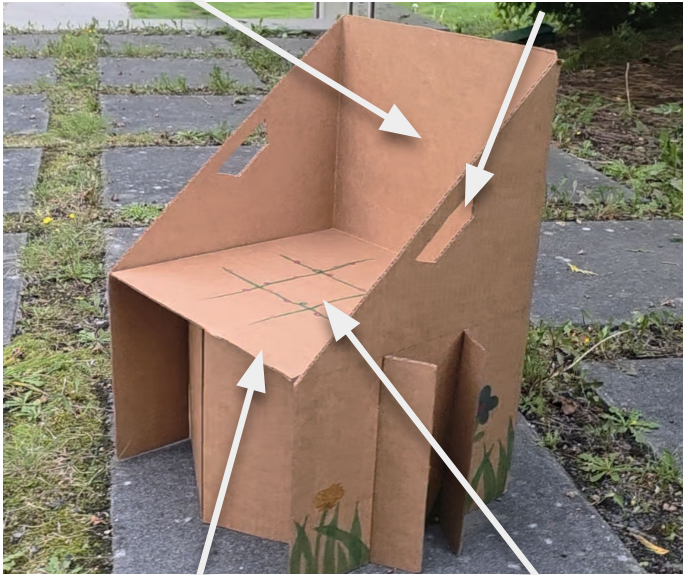


Figure 11: Final chair design, back view

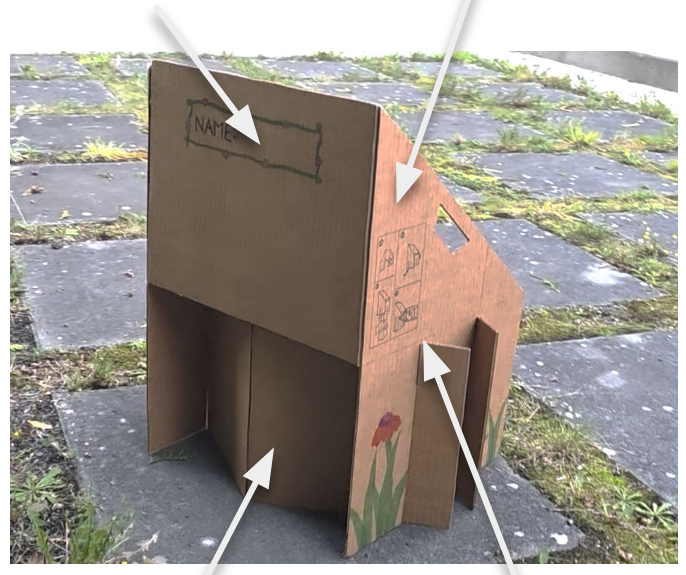


Figure 12: Final chair design, front view

Flip-up seat with hidden storage

Tic-tac-toe game board

Sturdy base (doubles as ring toss when placed upside down)

Clear, graphic instructions



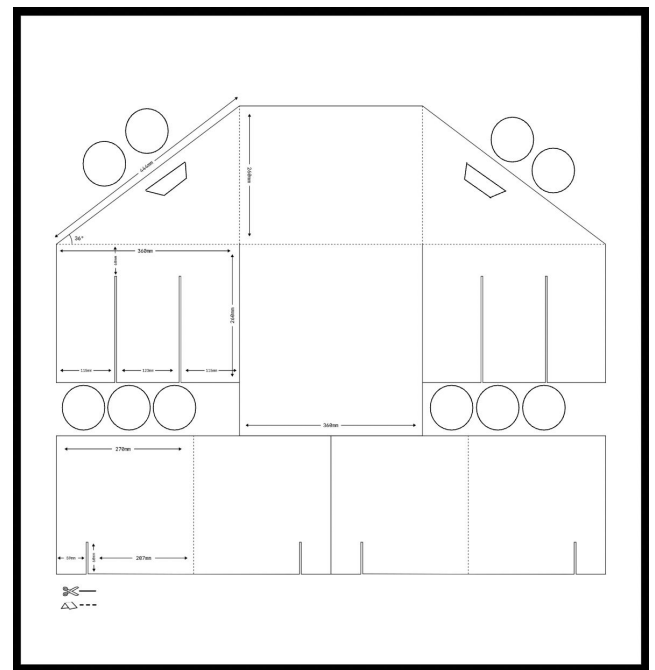
Fun playing pieces for ring toss and tic-tac-toe



Results

- Total rectangular area: 9,936 cm²
- Assembly time: 9.34 seconds
- Maximum weight support:
- Storage capacity: 18,000 cm³

Figure 13: Cutting pattern of Triangle Arm chair with absolute dimensions



Stages 4/5